Sensor Tow Stability

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1.0 OBJECTIVE

Screen 2–3 candidate sensor bodies, each at three ballast states, for aerodynamic stability on a tow line: no spinning, rotation or sustained oscillation. Measure the line angle to estimate sensor drag for the tow model (Script 5). Film every run for the proposal.

Rig. The university has no wind tunnel, and the rev A car tow is withdrawn for safety. This revision uses a leaf blower's jet, straightened in a short duct. Speed is read with an anemometer, so the same card works in a makeshift tunnel.

Limitation. The jet is slower and smaller than flight conditions, and a short line is not the in-flight pendulum. Stability is a screen and drag is semi-quantitative; the towed-flight test qualifies the sensor. A fin misalignment makes the body roll at a steady rate that rises in proportion to airspeed, so any rotation here would be faster in flight.

TEST AT A GLANCE

CREW	DURATION
3 people	**3–4** h
BODIES	BALLAST STATES
2–3	**3** each
DUCT	SITE
100–150 mm	**Indoor**

2.0 RULES THAT APPLY

- §3.1.1: the sensor must be "aerodynamically stable during deployment, operation and recovery", which the rules define as "remaining in a fixed orientation in the horizontal plane (not spinning or rotating) with no effect on the stability of the aircraft, at the discretion of the Flight Director."
- §3.1.1: minimum length 152.4 mm (6 in). Fins stay fixed (not removed for packing, not deployed), the external shape is the same for all missions, and ballast is internal.

3.0 TEST ARTICLES AND EQUIPMENT

- **Sensor bodies:** 2–3 fixed-geometry shapes from Sensor Systems (for example a ring tail, four fixed fins, or a fixed drag cone), at least 170 mm long for margin over the 152.4 mm minimum, each with an internal ballast bay. Paint a **contrasting longitudinal stripe** on each so any roll shows on video. Mark the tow point and CG.
- **Ballast:** light, mid and heavy internal states spanning the analysis range, weighed to 0.1 g on the bench scale (A01).
- **Blower** (PR-01 A07), clamped down, feeding a rigid 100 or 150 mm duct, 0.6–1.0 m long (A08). Drinking straws (A09) packed tight across the duct near its exit form the straightener.
- **Mount and line:** a rigid boom above the jet with an adjustable tie-off. Braided PE line (A11), 0.3–0.6 m long, runs to a ball-bearing swivel (A10) at the tow point, plus a secondary tether. The swivel lets a roll tendency show as rotation instead of being hidden by line twist.
- **Measurement:** vane anemometer (A03); a printed 10 mm grid behind the line, square to the side camera; two phones at 60 fps, side-on (line angle, pitch) and from above (roll stripe, yaw); air temperature and pressure (a phone app is enough); optionally the hanging scale (A02) at the tie-off for tension.

TABLE 01 Expected mean exit speed $V = Q/A$ (m/s) before straightener and duct losses. A smaller duct is faster but its uniform core is narrower; the jet survey gives the real values.

AIR VOLUME Q	Ø100 MM	Ø125 MM	Ø150 MM
10 m³/min	21.2	13.6	9.4
15 m³/min	31.8	20.4	14.1
20 m³/min	42.4	27.2	18.9

4.0 SAFETY

- Eye protection within 3 m of the jet; hearing protection beside the blower (A05, A06).
- Nobody stands downstream. Keep 5 m behind the body clear: a body that breaks free leaves at jet speed.
- Check the tether, knots, swivel and tie-off before each run. Stop the blower before touching anything in the jet.
- Route the cord away from the jet. Run indoors or sheltered, because outdoor wind disturbs the jet.

5.0 PROCEDURE

1. Photograph, weigh and measure each body and ballast state (length, diameter, fin span, mass, tow point, CG) and log them.
2. **Jet survey**, no body fitted. At the body station, about one duct diameter past the exit, traverse the anemometer across the jet horizontally and vertically in 15–20 mm steps. The **core** is where the speed is within 5 % of the centre value. The vane averages over its own diameter, so the core edge is approximate. Use Table 01 to pick the duct, and repeat for each blower setting.
3. Hang the body and adjust the tie-off until the whole body, fins included, sits inside the core at speed.
4. For each body, ballast state and setting, hold 30 s steady with both cameras running. Classify the run: **S** stable, **O** oscillating (amplitude, period), **C** coning, **R** rotating (fail). Log roll as none, intermittent or continuous.
5. At the top setting, flick the line once above the jet with a stick, hands outside the jet. Time the decay from the video.
6. Read the line angle θ from horizontal in three or more side-camera frames, plus the tension T if fitted. Log air temperature and pressure.
7. Repeat one run of the best body with the fixed attachment planned for flight in place of the swivel.

6.0 DATA REDUCTION

With the line θ above horizontal, tension T, weight W and lift L:

$$D = T\cos\theta, \qquad W - L = T\sin\theta, \qquad C_D A = \frac{2D}{\rho V^2}. \quad (1)$$

Without a tension reading, take $L \approx 0$, so $D \approx W\cot\theta$. Air density is $\rho = p/(R\,T_{\text{air}})$ with $R = 287.05$ J/(kg K), p in Pa (1 hPa = 100 Pa) and T_{air} in K. A near-vertical line amplifies angle errors, since $\Delta D/D = \Delta\theta/(\sin\theta\cos\theta)$: a 1° misreading moves D about 10 % at 80° and 20 % at 85°. Average the frames.

7.0 PASS CRITERIA AND USE OF RESULTS

Pass: no rotation (R) at any setting or ballast state, and oscillation after the flick dies out within about 2 s. Passing bodies go to the towed-flight test.

Use: the shape, tow point and ballast range define SR-08 (draft Table 04). The measured $C_D A$ replaces the assumed sensor drag in Script 5, and the best run is the proposal photo. File the sheet as `testing/data/TC-01_<date>.csv` and the media in `testing/media/TC-01/`.

8.0 DATA SHEET

Date _____ Site _____ Crew _____

Blower and setting _____ Duct Ø (mm) _____ Straightener length (mm) _____

Air temperature _____ Pressure (hPa) _____ Video file IDs _____

TABLE 02 Jet survey at the body station. Core: within 5 % of the centre speed.

OFFSET, MM	HORIZ., M/S	VERT., M/S
−60		
−45		
−30		
−15		
0		
+15		
+30		
+45		
+60		

TABLE 03 Test articles. Masses to 0.1 g.

BODY	SHAPE, LENGTH, Ø, FIN SPAN	LIGHT, G	MID, G	HEAVY, G
A				
B				
C				

TABLE 04 Run log. Behaviour: S stable, O oscillating, C coning, R rotating (fail). Roll: none, intermittent, continuous.

RUN	BODY	BALLAST, G	SETTING	V, M/S	θ, DEG	T, G	BEHAVIOUR	ROLL	DECAY, S	NOTES
1										
2										
3										
4										
5										
6										
7										
8										
9										
10										
11										
12										
13										
14										
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16										